

A CONVERSE COMPARISON THEOREM FOR BSDES AND RELATED PROPERTIES OF g -EXPECTATION

Question: Given two generators g_1 and g_2 .
 $\forall \xi \in L_T^2, \mathcal{E}_{g_1}(\xi) \leq \mathcal{E}_{g_2}(\xi) \stackrel{?}{\Rightarrow} g_1 \leq g_2$.

Notation:

$(\Omega, \mathcal{F}, P)$ - Probability Space

$(W_t)_{t \geq 0}$ - Standard d -dimensional B.M.

$(\mathcal{F}_t)_{t \geq 0}$ - σ -algebra generated by $(W_t)_{t \geq 0}$

(IP-augmentation = right-continuous and complete)

$\mathcal{S}_2 = \{\psi \text{ progressively measurable; } \|\psi\|_{\mathcal{S}_2}^2 := E[\sup_{0 \leq t \leq T} |\psi_t|^2] < \infty\}$

$\mathcal{H}_2 = \{\psi \text{ progressively measurable; } \|\psi\|_{\mathcal{H}_2}^2 := E[\int_0^T \|\psi_t\|^2 dt] < \infty\}$

$(g(t, y, z))_{t \in [0, T]} \subset \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d$.
 - progressively measurable

(A1) P -a.s. $\forall t, \forall (y, z), \forall (y', z')$

$$|g(t, y, z) - g(t, y', z')| \leq K(|y - y'| + |z - z'|)$$

(A2) $(g(t, 0, 0))_{t \in [0, T]} \in \mathcal{H}_2$.

(A3) P -a.s., $\forall (t, y), g(t, y, 0) = 0$

(A4) P -a.s., $\forall (y, z), t \mapsto g(t, y, z)$ is continuous.

It is by now well known that under the assumptions (A1) and (A2), for any random variable ξ in $L^2(\mathcal{F}_T)$, the BSDE

$$Y_t = \xi + \int_t^T g(s, Y_s, Z_s) ds - \int_t^T Z_s \cdot dW_s, \quad 0 \leq t \leq T, \quad (1)$$

has a unique adapted solution, say $((Y_t, Z_t))_{t \in [0, T]}$ s.t. Z is in the space $\mathcal{H}_2$. Actually, by classical results in this field, the process Y belongs to $\mathcal{S}_2$.

By (A1) and Ineq: $K|y| \cdot \|Z\| \leq K^2|y|^2 + \frac{1}{2}\|Z\|^2$

$$\begin{aligned} e^{\beta u} |Y_u|^2 + \int_u^T e^{\beta s} \|Z_s\|^2 ds &\leq e^{\beta T} |\xi|^2 + 2 \int_u^T e^{\beta s} |g(s, Y_s, Z_s)| ds - \int_u^T \beta e^{\beta s} |Y_s|^2 ds - \mathcal{M}_u \\ &\leq e^{\beta T} |\xi|^2 + 2 \int_u^T e^{\beta s} |g(s, Y_s, Z_s) - g(s, 0, 0)| ds + \int_u^T \beta e^{\beta s} |Y_s|^2 ds - \mathcal{M}_u \\ &\leq e^{\beta T} |\xi|^2 + 2 \int_u^T e^{\beta s} K(|Y_s| \|Z_s\|) ds + \int_u^T \beta e^{\beta s} |Y_s|^2 ds - \mathcal{M}_u \\ &\leq e^{\beta T} |\xi|^2 + \int_u^T \beta e^{\beta s} |Y_s|^2 ds - \mathcal{M}_u \\ &\leq e^{\beta T} |\xi|^2 + \int_u^T e^{\beta s} [2K(K + K^2 - \beta) |Y_s|^2 + \frac{1}{2} e^{\beta s} \|Z_s\|^2] ds \\ &\quad + 2 \int_u^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds - \mathcal{M}_u. \end{aligned}$$

We deduce that

$$e^{\beta u} |Y_u|^2 + \frac{1}{2} \int_u^T e^{\beta s} \|Z_s\|^2 ds \leq e^{\beta T} |\xi|^2 + 2 \int_u^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds - 2 \int_u^T e^{\beta s} Y_s Z_s \cdot dW_s \quad (2)$$

Taking C.E w.r.t. $\mathcal{F}_t$. ($u \leftrightarrow t$)

$$E[\int_t^T e^{\beta s} \|Z_s\|^2 ds | \mathcal{F}_t] \leq 2E[e^{\beta T} |\xi|^2 + 2 \int_t^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds | \mathcal{F}_t] \quad (3)$$

From (2):

$$\sup_{t \leq u \leq T} e^{\beta u} |Y_u|^2 \leq e^{\beta T} |\xi|^2 + 2 \int_t^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds + 4 \sup_{t \leq u \leq T} \int_t^u e^{\beta s} Y_s Z_s \cdot dW_s$$

By BDG Ineq. (C is universally const.)

$$E[\sup_{t \leq u \leq T} e^{\beta u} |Y_u|^2 | \mathcal{F}_t] \leq E[e^{\beta T} |\xi|^2 + 2 \int_t^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds | \mathcal{F}_t] + CE[E(\int_t^T e^{2\beta s} |Y_s|^2 \|Z_s\|^2 ds) | \mathcal{F}_t].$$

$$\text{BDG Ineq: } \forall p > 0, \exists b_p, B_p, b_p E[\langle M \rangle_T^p] \leq E[(\sup_{0 \leq t \leq T} |M_t|)^p] \leq B_p E[\langle M \rangle_T^p].$$

By Ineq: $ab \leq \frac{a^2}{2} + \frac{b^2}{2}$.

$$E[\sup_{t \leq u \leq T} e^{\beta u} |Y_u|^2 | \mathcal{F}_t] \leq E[e^{\beta T} |\xi|^2 + 2 \int_t^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds | \mathcal{F}_t] + \frac{C^2}{2} E[\int_t^T e^{\beta s} \|Z_s\|^2 ds | \mathcal{F}_t] + \frac{1}{2} E[\sup_{t \leq u \leq T} e^{\beta u} |Y_u|^2 | \mathcal{F}_t]$$

Combining (3) with $\uparrow$:

$$E[\sup_{t \leq u \leq T} e^{\beta u} |Y_u|^2 + \int_t^T e^{\beta s} \|Z_s\|^2 ds | \mathcal{F}_t] \leq CE[e^{\beta T} |\xi|^2 + 2 \int_t^T e^{\beta s} |Y_s| |g(s, 0, 0)| ds]$$

Z. CHEN's Result:

Theorem 2.1 Let the assumptions (A1), (A3) and (A4) hold for g_1 and g_2 and let us assume moreover that, for each $\xi \in L^2(\mathcal{F}_T)$, $\mathcal{E}_{g_1}(\xi) = \mathcal{E}_{g_2}(\xi)$. Then, P -a.s., $\forall (t, y, z), g_1(t, y, z) = g_2(t, y, z)$.

A Comment of (A3):

ξ - $\mathcal{F}_s$ Measurable ($s < T$).

$\mathcal{E}_g(\xi | \mathcal{F}_t) = \xi$ if $s \leq t \leq T$ and $\mathcal{E}_g(\xi | \mathcal{F}_t) = y_t$ for $0 \leq t < s$.

$((Y_r, Z_r))_{r \in [0, s]}$ - solution of BSDE:

$$Y_r = \xi + \int_r^s g(u, Y_u, Z_u) du - \int_r^s Z_u \cdot dW_u, \quad 0 \leq r \leq s.$$

Technical Results.

Prop 2.2

$\xi \in L^2(\mathcal{F}_T)$. g - (A1) (A2)

If $((Y_t, Z_t))_{t \in [0, T]}$ - BSDE(1), we have for $\beta = 2(K + K^2)$

$$E[\sup_{t \leq s \leq T} e^{\beta s} |Y_s|^2 + \int_t^T e^{\beta s} \|Z_s\|^2 ds | \mathcal{F}_t] \leq CE\{\xi^2 + (\int_t^T e^{\frac{\beta s}{2}} |g(s, 0, 0)| ds)^2 | \mathcal{F}_t\}. \quad (C \text{ is a const.})$$

Proof. Itô Lemma $\Rightarrow (M_u = 2 \int_u^T e^{\beta s} Y_s Z_s \cdot dW_s)$

$$e^{\beta u} |Y_u|^2 + \int_u^T e^{\beta s} \|Z_s\|^2 ds = e^{\beta T} |\xi|^2 + 2 \int_u^T e^{\beta s} Y_s \cdot g(s, Y_s, Z_s) ds - \int_u^T \beta e^{\beta s} |Y_s|^2 ds - M_u.$$

(Why: Consider $f(t, y) = e^{\beta t} |y|^2$, $df_t = -g dt + Z_t \cdot dW_t$
 $df(t, Y_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial y}(t, Y_t) dY_t + \frac{1}{2} \frac{\partial^2 f}{\partial y^2} d\langle Y \rangle_t$
 $d\langle Y \rangle_t = \|Z_t\|^2 dt, \dots$)

With the fact we finish the proof:

$$CE[\int_t^T e^{\beta s} |Y_s|^2 | \mathcal{F}_t] \leq \frac{1}{2} E[\sup_{t \leq u \leq T} e^{\beta u} |Y_u|^2 | \mathcal{F}_t] + \frac{C^2}{2} E[(\int_t^T e^{\frac{\beta s}{2}} |g(s, 0, 0)| ds)^2 | \mathcal{F}_t] \quad \square$$

(A5). Let $b: \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $\sigma: \mathbb{R}^n \rightarrow \mathbb{R}^{n \times d}$ be two Lipschitz functions.

We denote by $X^{t,x}$ the solution of the SDE:

$$X_s^{t,x} = x + \int_t^s b(X_u^{t,x}) du + \int_t^s \sigma(X_u^{t,x}) \cdot dW_u, \quad s \geq t, \quad (4)$$

with the usual convention $X_s^{t,x} = x$ if $s < t$.

Let $g: \Omega \times [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R}$ be a function s.t. the stochastic process $(g(s, y, z))_{s \in [0, T]}$ is progressively measurable for each (y, z) . Fix $(t, x, y, p) \in [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n$ and let us denote by $((Y_s^{t,x,y,p}, Z_s^{t,x,y,p}))_{s \in [0, t+\varepsilon]}$, for $\varepsilon > 0$ small enough, the solution of the BSDE on $[0, t+\varepsilon]$:

$$Y_s^{t,x,y,p} = y + p \cdot (X_{t+\varepsilon}^{t,x} - x) + \int_s^{t+\varepsilon} g(u, Y_u^{t,x,y,p}, Z_u^{t,x,y,p}) du - \int_s^{t+\varepsilon} Z_u^{t,x,y,p} \cdot dW_u. \quad (5)$$

Prop 2.3

g - (A1) (A2) (A4). The notation (A5) holds.

Assume $E[\sup_{0 \leq t \leq T} |g(t, 0, 0)|^2]$ is finite.

For each $(t, x, y, p) \in [0, T] \times \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n$,

$$L^2 - \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \int Y_t^{t,x,y,p} - y = g(t, y, \sigma^t(x)p) + p \cdot b(x).$$

Proof. Fixed (t, x, y, p) and then write $(Y^\varepsilon, Z^\varepsilon)$ instead.

Pick $\varepsilon > 0$, for $t \leq s \leq t + \varepsilon$.

$$\tilde{Y}_s^\varepsilon = Y_s^\varepsilon - (y + p \cdot (X_{t+\varepsilon}^{t,x} - x)). \quad \tilde{Z}_s^\varepsilon = Z_s^\varepsilon - \sigma^t(X_{t+\varepsilon}^{t,x})p.$$

Using Itô's formula: on $[t, t+\varepsilon]$, $(\tilde{Y}^\varepsilon, \tilde{Z}^\varepsilon)$ solves

$$\text{BSDE: } \tilde{Y}_s^\varepsilon = \int_s^{t+\varepsilon} g(u, y + p \cdot (X_{t+\varepsilon}^{t,x} - x) + \tilde{Y}_u^\varepsilon, \sigma^t(X_{t+\varepsilon}^{t,x})p + \tilde{Z}_u^\varepsilon) du + \int_s^{t+\varepsilon} p \cdot b(X_u^{t,x}) du - \int_u^{t+\varepsilon} \tilde{Z}_u^\varepsilon \cdot dW_u. \quad (6)$$

By Prop 2.2.

$$E[\sup_{t \leq s \leq t+\varepsilon} |\tilde{Y}_s^\varepsilon|^2 + \int_t^{t+\varepsilon} \|\tilde{Z}_s^\varepsilon\|^2 ds | \mathcal{F}_t] \\ \leq C e^{2(K+K^*)\varepsilon} E[(\int_t^{t+\varepsilon} |g(u, y, p \cdot (x_s^{t,x} - x), \sigma^t(x_u^{t,x})p) \\ + p \cdot b(x_u^{t,x})| du)^2 | \mathcal{F}_t]$$

Since g, b, σ - Lipschitz.

$$E[\sup_{t \leq s \leq t+\varepsilon} |\tilde{Y}_s^\varepsilon|^2 + \int_t^{t+\varepsilon} \|\tilde{Z}_s^\varepsilon\|^2 ds | \mathcal{F}_t] \\ \leq C_{x,y,p} \varepsilon^2 E[1 + \sup_{t \leq s \leq t+\varepsilon} (|x_s^{t,x}|^2 + |g(s, 0, 0)|^2) | \mathcal{F}_t].$$

Additional Assumption $\Rightarrow E[\sup_{0 \leq s \leq T} |x_s^{t,x}|^2 + |g(s, 0, 0)|^2] \leq C(1 + |x|^2)$

Estimate:

$$E[\sup_{t \leq s \leq t+\varepsilon} |\tilde{Y}_s^\varepsilon|^2 + \int_t^{t+\varepsilon} \|\tilde{Z}_s^\varepsilon\|^2 ds] \leq C\varepsilon^2 \quad (7)$$

Taking CE in BSDE (6)

$$\frac{1}{\varepsilon} (Y_t^\varepsilon - y) = \frac{1}{\varepsilon} \tilde{Y}_t^\varepsilon = \frac{1}{\varepsilon} E[\int_t^{t+\varepsilon} \dots du | \mathcal{F}_t]$$

Split the right side:

$$\frac{1}{\varepsilon} (Y_t^\varepsilon - y) = \frac{1}{\varepsilon} E[\int_t^{t+\varepsilon} |g(u, y, p \cdot (x_s^{t,x} - x), \sigma^t(x_u^{t,x})p) \\ + p \cdot b(x_u^{t,x})| du | \mathcal{F}_t] + R_\varepsilon.$$

$$R_\varepsilon := \frac{1}{\varepsilon} E[\int_t^{t+\varepsilon} \{g(u, y, p \cdot (x_u^{t,x} - x)) + \tilde{Y}_u^\varepsilon, \tilde{Z}_u^\varepsilon + \sigma^t(x_u^{t,x})p\} \\ - g(u, y, p \cdot (x_u^{t,x} - x), \sigma^t(x_u^{t,x})p) \} du | \mathcal{F}_t]$$

By g -Lipschitz.

$$|R_\varepsilon| \leq \frac{K}{\varepsilon} E[\int_t^{t+\varepsilon} (|\tilde{Y}_u^\varepsilon| + \|\tilde{Z}_u^\varepsilon\|) du | \mathcal{F}_t]$$

By Hölder Ineq. and Ineq: $(x+y)^2 \leq 2(x^2 + y^2)$

$$E[|R_\varepsilon|^2] \leq \frac{2K^2}{\varepsilon} E[\int_t^{t+\varepsilon} (|\tilde{Y}_u^\varepsilon|^2 + \|\tilde{Z}_u^\varepsilon\|^2) du]$$

$$\leq C\varepsilon \quad (C \text{ is universal const.})$$

$$\Rightarrow E[|R_\varepsilon|^2] \rightarrow 0 \Rightarrow R_\varepsilon \rightarrow 0 \quad (L^2).$$

Remark. As we can see in the proof, the continuity of the process $(g(t, y, z))_{t \in [0, T]}$ (assumption (A4)) is not really needed. We can prove the result if this process is only right-continuous.

Remark. It is worth noting that the assumption " $E[\sup_{0 \leq t \leq T} |g(t, 0, 0)|^2]$ finite" holds when g satisfies (A3) or in the Markovian situation (see the subsection 3.2 below).

The deterministic case.

• Main Result.

$$g: [0, T] \times \mathbb{R} \times \mathbb{R}^d \rightarrow \mathbb{R} \quad (A1) (A3)$$

Quest: If ξ is independent of $\mathcal{F}_t$, $E_g(\xi | \mathcal{F}_t) \stackrel{?}{=} E_g(\xi)$.

Prop 3.1

g - deterministic, for each $\xi \in L^2(\mathcal{F}_T)$.

$E_g(\xi | \mathcal{F}_t) = E_g(\xi)$ as soon as ξ is dependent of $\mathcal{F}_t$

Proof. $E_g(\xi) = E_g(E_g(\xi | \mathcal{F}_t))$ and by (A3), $E_g(c) = c$

Shift method $\rightarrow E_g(\xi | \mathcal{F}_t)$ is deterministic

$\forall 0 \leq s \leq T-t, W'_s = W_{t+s} - W_t, \{W'_s, 0 \leq s \leq T-t\}$ is a B.M. w.r.t. $\mathcal{F}'_s = \mathcal{F}_{t+s}^t$ (σ -algebra generated by increments of B.M. W after t .)

$((Y'_s, Z'_s))_{0 \leq s \leq T-t}$ - BSDE:

$$Y'_s = \xi + \int_s^{T-t} g(t+u, Y'_u, Z'_u) du - \int_s^{T-t} Z'_u dW'_u, \quad 0 \leq s \leq T-t.$$

$$Y'_{T-t} = \xi + \int_{T-t}^{T-t} g(t+u, Y'_u, Z'_u) du - \int_{T-t}^{T-t} Z'_u dW'_u, \quad t \leq u \leq T.$$

$$Y'_{t+u} = \xi + \int_{t+u}^{T-t} g(r, Y'_r, Z'_r) dr - \int_{t+u}^{T-t} Z'_r dW'_r, \quad t \leq u \leq T.$$

$$= \xi + \int_{t+u}^{T-t} g(r, Y'_r, Z'_r) dr - \int_{t+u}^{T-t} Z'_r dW_r, \quad t \leq u \leq T.$$

By Uniqueness of the solution of BSDE:

$$((Y'_{t+u}, Z'_{t+u}))_{u \in [0, T-t]} = ((Y'_u, Z'_u))_{u \in [0, T-t]}$$

$$\Rightarrow Y'_0 = Y_t. \text{ since } Y'_0 \text{ is deterministic}$$

The proof is complete. $\square$. $\hookrightarrow Y'_0$ is independent of $\mathcal{F}_t$. But also $\mathcal{F}_t$ -measurable.

$$\text{Set } Q_\varepsilon = \frac{1}{\varepsilon} E[\int_t^{t+\varepsilon} \{g(u, y, p \cdot (x_u^{t,x} - x), \sigma^t(x_u^{t,x})p) \\ + p \cdot b(x_u^{t,x}) - b(x) \} du | \mathcal{F}_t]$$

$$P_\varepsilon = \frac{1}{\varepsilon} E[\int_t^{t+\varepsilon} \{g(u, y, \sigma^t(x)p) - g(t, y, \sigma^t(x)p) \} du | \mathcal{F}_t]$$

Now we will prove $Q_\varepsilon \rightarrow 0$ in L^2 .

By g, b, σ - Lipschitz.

$$|Q_\varepsilon| \leq \frac{C}{\varepsilon} E[\int_t^{t+\varepsilon} |x_u^{t,x} - x| du | \mathcal{F}_t]$$

By Hölder.

$$E|Q_\varepsilon|^2 \leq \frac{C^2}{\varepsilon} E[\int_t^{t+\varepsilon} |x_u^{t,x} - x|^2 du | \mathcal{F}_t]$$

Since (H. KUNITA) we have,

$$\forall r \geq 1, E[\sup_{0 \leq s \leq T} |x_s^{t,x}|^r] \leq C_T(1 + |x|^r)$$

$\Rightarrow u \mapsto E[|x_u^{t,x} - x|^2]$ continuous.

equals to 0 at time t .

$\Rightarrow Q_\varepsilon \rightarrow 0 \quad (L^2)$

Finally. By Hölder Ineq:

$$E|P_\varepsilon|^2 \leq E[\int_t^{t+\varepsilon} |g(u, y, \sigma^t(x)p - g(t, y, \sigma^t(x)p)|^2 du]$$

By (A4),

$$\frac{1}{\varepsilon} \int_t^{t+\varepsilon} |g(u, y, \sigma^t(x)p - g(t, y, \sigma^t(x)p)|^2 du \rightarrow 0 \quad P\text{-a.s.}$$

Moreover,

$$\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \dots du \leq C(\sup_{0 \leq s \leq T} |g(s, 0, 0)|^2 + |y|^2 + \|\sigma^t(x)p\|^2)$$

- Integrable.

$$P_\varepsilon \rightarrow 0 \quad (L^2)$$

$$R_\varepsilon, Q_\varepsilon, P_\varepsilon \rightarrow 0 \quad (L^2) + \text{Lebesgue DCT} \Rightarrow \text{Prop 2.3} \quad \square.$$

Remark. Actually, if for each random variable $\xi \in L^2(\mathcal{F}_T)$, we have $E_g(\xi | \mathcal{F}_t) = E_g(\xi)$ as soon as ξ is independent of $\mathcal{F}_t$ then g is deterministic.

Indeed, for any (t, y, z) , we have, by Proposition 2.3,

$$g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \{E_g(y + z \cdot (W_{t+\varepsilon} - W_t) | \mathcal{F}_t) - y\} = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \{E_g(y + z \cdot (W_{t+\varepsilon} - W_t)) - y\},$$

and thus g is deterministic.

EXAMPLE (g is not deterministic)

Fix $T > 0$, Pick $t \in (0, T)$, $f: \mathbb{R} \rightarrow \mathbb{R}$ continuous, bounded.

Def: For each $(s, z) \in [0, T] \times \mathbb{R}$, $g(s, z) := f(W_{s+t})z$.

Set $\xi = W_T - W_t$ (independent of $\mathcal{F}_t$). (By KPQ Prop 2.2)

$$E_g(\xi | \mathcal{F}_t) = E(\xi \exp(\int_t^T f(W_{s+t}) dW_s - \frac{1}{2} \int_t^T f^2(W_{s+t}) ds) | \mathcal{F}_t)$$

$$= E_Q((W_T - W_t) | \mathcal{F}_t) \quad (\text{By Girsanov})$$

where Q is probability measure on $(\Omega, \mathcal{F}_T)$ whose density is

$$Z_T := \exp(\int_0^T f(W_{s+t}) dW_s - \frac{1}{2} \int_0^T f^2(W_{s+t}) ds)$$

$\{\tilde{W}_r = W_r - \int_0^r f(W_{s+t}) ds, 0 \leq r \leq T\}$ is a B.M.

$$\Rightarrow W_T - W_t = \tilde{W}_T - \tilde{W}_t + \int_t^T f(W_{s+t}) ds$$

$$= \tilde{W}_T - \tilde{W}_t + (T-t)f(W_t)$$

$$\Rightarrow E_g(\xi | \mathcal{F}_t) = (T-t)f(W_t) \rightarrow \text{Not detm. if } f \text{ not a const.}$$

Theorem 3.2

$$g_i - (A1) (A3) (A4) \quad i=1, 2.$$

$$\forall \xi \in L^2(\mathcal{F}_T), E_g(\xi) \leq E_{g_2}(\xi) \Rightarrow$$

$$\forall t \in [0, T], \forall (y, z) \in \mathbb{R} \times \mathbb{R}^d, g_1(t, y, z) \leq g_2(t, y, z).$$

Proof. Consider $\xi_n = y + z \cdot (W_{t+\frac{1}{n}} - W_t)$ ($n \in \mathbb{N}^+$, large enough)

$$(A3) \Rightarrow E[\sup_{0 \leq s \leq T} |g(s, 0, 0)|^2] \text{ finite} \xrightarrow{\text{Prop 2.3}}$$

Fix y, z .

$$n(E_g(\xi_n | \mathcal{F}_t) - y) \rightarrow g_i(t, y, z), \text{ as } n \rightarrow \infty. \quad (L^2).$$

$$\xi_n \text{ independent of } \mathcal{F}_t. E_{g_i}(\xi_n | \mathcal{F}_t) = E_{g_i}(\xi_n) \quad (i=1, 2)$$

$$\Rightarrow E_{g_1}(\xi_n | \mathcal{F}_t) = E_{g_1}(\xi_n) \leq E_{g_2}(\xi_n) = E_{g_2}(\xi_n | \mathcal{F}_t) \Rightarrow \text{Prop 3.2} \quad \square$$

• An application of PDEs.

Let $f : \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^d \longrightarrow \mathbb{R}$ be a function s.t.

(A 6). There exist two constants $K \geq 0$ and $q \geq 1$, s.t.

- $\forall (y, z), \quad x \longrightarrow f(x, y, z)$ is continuous;
- $\forall x, \forall (y, z), (y', z'), \quad |f(x, y, z) - f(x, y', z')| \leq K(|y - y'| + \|z - z'\|)$;
- $\forall x, \quad |f(x, 0, 0)| \leq K(1 + |x|^q)$.

In addition, let us consider $h : \mathbb{R}^n \longrightarrow \mathbb{R}$ a continuous function with polynomial growth. Then it is by now well known that BSDEs in Markovian context give a nonlinear Feynman-Kac formula for semilinear PDEs in the sense that the viscosity solution, say u , of the semilinear PDE

$$\partial_t u = \frac{1}{2} \text{tr}(\sigma \sigma^t(x) D^2 u) + b(x) \cdot \nabla u + f(x, u, \sigma^t(x) \nabla u), \quad (t, x) \in \mathbb{R}_+ \times \mathbb{R}^n, \quad u(0, \cdot) = h(\cdot), \quad (8)$$

is equal, for each (t, x) , to $Y_0^{t,x}$ where $((Y_s^{t,x}, Z_s^{t,x}))_{s \in [0,t]}$ is the solution of the BSDE

$$Y_s^{t,x} = h(X_s^{0,x}) + \int_s^t f(X_u^{0,x}, Y_u^{t,x}, Z_u^{t,x}) du - \int_s^t Z_u^{t,x} \cdot dW_u, \quad 0 \leq s \leq t,$$

where $X^{0,x}$ is the solution of the SDE (4).

Prop 3.3

f_1, f_2 - (A6). and b, σ - (A5)

If for $(x, y, p) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n$.

$\exists \eta > 0$. s.t. $\forall \varepsilon < \eta, u_1(\varepsilon, x) \leq u_2(\varepsilon, x)$.

where u_i - viscosity solution of PDE (8).

with semilinear part f_i . and

initial condition $h(\cdot) = y + p \cdot (-x)$.

Then $f_1(x, y, \sigma^t(x)p) \leq f_2(x, y, \sigma^t(x)p)$.

By Prop 2.3. $n(Y_t^i(\xi_n) - y) \rightarrow g_i(t, y, z)$ (L^2)

By the hypothesis. $n(Y_t^1(\xi_n) - y) \leq n(Y_t^2(\xi_n) - y)$

↓. Extracting a subsequence to get the convergence $\mathbb{P} - a.s.$ and then passing to the limit when n goes to infinity, we obtain, the inequality $\mathbb{P} - a.s., g_1(t, y, z) \leq g_2(t, y, z)$. By continuity, we obtain finally that $\mathbb{P} - a.s.,$

$$\forall (t, y, z) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d, \quad g_1(t, y, z) \leq g_2(t, y, z).$$

The proof is complete. $\square$

Remark. We give in this remark the main lines of a different proof of this result. This approach is based on a "nonlinear decomposition theorem of Doob-Meyer's type" due to S. PENG [9]. For a given ξ , the assumption of the theorem says that the process $Y^2(\xi)$ is a g_1 -supermartingale and actually it can be seen that it is a g_1 -supermartingale in the strong sense. Therefore, we apply Theorem 3.3 in [9] to see that this process is also a g_1 -supersolution. From this we deduce easily that, for each ξ and for each $0 \leq t_0 < t \leq T$,

$$\frac{1}{t - t_0} \int_{t_0}^t \mathbb{E} \left\{ g_1(u, Y_u^2(\xi), Z_u^2(\xi)) \right\} du \leq \frac{1}{t - t_0} \int_{t_0}^t \mathbb{E} \left\{ g_2(u, Y_u^2(\xi), Z_u^2(\xi)) \right\} du,$$

and choosing, as in Z. CHEN [1], $\xi = X_T$, where, for a given (t_0, y_0, z_0) , X is the solution of the SDE

$$X_t = y_0 - \int_{t_0}^t g_2(u, X_u, z_0) du + \int_{t_0}^t z_0 \cdot dW_u,$$

we obtain, letting $t \longrightarrow t_0^+$, $g_1(t_0, y_0, z_0) \leq g_2(t_0, y_0, z_0)$.

• y-independent case.

$g : \Omega \times [0, T] \times \mathbb{R}^d \rightarrow \mathbb{R}$. & (A1) (A3)

Lem 4.2

$\forall (\xi, \eta) \in L^2(\mathcal{F}_T) \times L^2(\mathcal{F}_t), \forall t \leq s \leq T:$

$$\mathbb{E} g(\xi - \eta | \mathcal{F}_t) = \mathbb{E}(\xi | \mathcal{F}_t) - \eta.$$

Particularly, $\forall 0 \leq s \leq t \leq T, \mathbb{E} g[\xi - \mathbb{E} g(\xi | \mathcal{F}_t) | \mathcal{F}_t] = 0$.

Lem 4.3

g - (A1) (A2) (A3)

$\forall \xi \in L^2(\mathcal{F}_T), \forall x \in \mathbb{R}, \forall t \in [0, T]$

$\mathbb{E} g(\xi - x | \mathcal{F}_t) = \mathbb{E} g(\xi | \mathcal{F}_t) - x \Rightarrow g$ not id. of y .

Proof. Fix $(x, y, p) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n$.

$\exists \eta > 0. \forall \varepsilon > \eta. {}^1 Y_0^{\varepsilon, x} \leq {}^2 Y_0^{\varepsilon, x}$.

$({}^1 Y^{\varepsilon, x}, {}^1 Z^{\varepsilon, x})$ - BSDE:

$${}^i Y_s^{\varepsilon, x} = y + p \cdot (X_s^{0, x} - x) + \int_s^T f_i(X_u^{0, x}, {}^i Y_u^{\varepsilon, x}, {}^i Z_u^{\varepsilon, x}) du - \int_s^T {}^i Z_u^{\varepsilon, x} dW_u, \quad 0 \leq s \leq T.$$

By SDE's result: $\mathbb{E} \sup_{0 \leq s \leq \eta} |X_s^{0, x}|^{2r} \leq C_\eta (1 + |x|^{2r})$

and the growth of $f: \mathbb{E} \sup_{0 \leq s \leq \eta} |f(X_s^{0, x}, 0, 0)|^2$ finite.

The assumption yields:

$$\forall \varepsilon < \eta, \frac{1}{\varepsilon} \{ {}^1 Y_0^{\varepsilon, x} - y \} = \frac{1}{\varepsilon} \{ {}^2 Y_0^{\varepsilon, x} - y \}$$

By Prop 2.3.

$$\frac{1}{\varepsilon} \{ {}^i Y_0^{\varepsilon, x} - y \} \xrightarrow{\varepsilon \rightarrow 0^+} f_i(x, y, \sigma^t(x)p) + p \cdot b(x) \quad (L^2)$$

Moreover, $\frac{1}{\varepsilon} \{ {}^i Y_0^{\varepsilon, x} - y \}$ dtm. convergence holds in $\mathbb{R}$ which gives the result. $\square$

Converse comparison theorem and application.

• A converse to the comparison theorem for BSDEs

Thm. 4.1

g_i - (A1) (A3) (A4) ($i=1, 2$)

Assume $\forall \xi \in L^2(\mathcal{F}_T), \forall t \in [0, T], Y_t^1(\xi) \leq Y_t^2(\xi)$

Then P-a.s. $\forall t \in [0, T] \forall (y, z) \in \mathbb{R} \times \mathbb{R}^d$.

$$g_1(t, y, z) \leq g_2(t, y, z).$$

Proof. Fix $(t, y, z) \in [0, T] \times \mathbb{R} \times \mathbb{R}^d, n \in \mathbb{N}_+$ (large enough)

Consider $\xi_n = y + z \cdot (W_{t+\frac{1}{n}} - W_t)$

Proof. $g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \{ \mathbb{E} g(y + z \cdot (W_{t+\varepsilon} - W_t) | \mathcal{F}_t) - y \}$ (L^2).

$$\mathbb{E} g(y + z \cdot (W_{t+\varepsilon} - W_t) | \mathcal{F}_t) = y + \mathbb{E} g(z \cdot (W_{t+\varepsilon} - W_t) | \mathcal{F}_t)$$

$$g(t, y, z) = \lim_{\varepsilon \rightarrow 0^+} \frac{1}{\varepsilon} \mathbb{E} g[z \cdot (W_{t+\varepsilon} - W_t) | \mathcal{F}_t].$$

Hence g does not depend on y . $\square$

Thm 4.4

g_i - (A1) (A3) (A4) ($i=1, 2$)

$\forall \xi \in L^2(\mathcal{F}_T), \mathbb{E} g_1[\xi] \leq \mathbb{E} g_2[\xi]$ 与 Thm 4.1 不同版

$\Rightarrow$ P-a.s. $\forall s \in [0, T], \forall z \in \mathbb{R}^d, g_1(s, z) \leq g_2(s, z)$

Proof. $\forall \xi \in L^2(\mathcal{F}_T), t \in [0, T], \mathbb{E} g_1[\xi | \mathcal{F}_t] \leq \mathbb{E} g_2[\xi | \mathcal{F}_t]$

$\forall \zeta \in L^2(\mathcal{F}_T), t \in [0, T]$.

$$\mathbb{E} g_1[\zeta - \mathbb{E} g_1[\zeta | \mathcal{F}_t]] \leq \mathbb{E} g_2[\zeta - \mathbb{E} g_1[\zeta | \mathcal{F}_t]]$$

By Lem 4.2. $\mathbb{E} g_1[\zeta - \mathbb{E} g_1[\zeta | \mathcal{F}_t] | \mathcal{F}_t] = 0$

$$\mathbb{E} g_2[\zeta - \mathbb{E} g_1[\zeta | \mathcal{F}_t]] = \mathbb{E} g_2[\mathbb{E} g_2[\zeta - \mathbb{E} g_1[\zeta | \mathcal{F}_t] | \mathcal{F}_t]]$$

$$\text{这些结论在硕博论文中可以找到} \sim \mathbb{E} g_2[\mathbb{E} g_2[\zeta | \mathcal{F}_t] - \mathbb{E} g_1[\zeta | \mathcal{F}_t]] \geq 0$$

Fix $\xi \in L^2(\mathcal{F}_T), t \in [0, T]$. Let $\zeta = \xi 1_{\eta < 0}$. $\eta = 1$.

$$\mathbb{E} g_1(\xi 1_{\eta < 0} | \mathcal{F}_t) = 1_{\eta < 0} \mathbb{E} g_1(\xi | \mathcal{F}_t) \quad (i=1, 2)$$

$$\Rightarrow \mathbb{E} g_2(\eta 1_{\eta < 0}) \geq 0$$

But $\eta 1_{\eta < 0}$ nonpositive $\Rightarrow \mathbb{E} g_1(\eta 1_{\eta < 0}) \leq \mathbb{E} g_2[0] = 0$

$$\Rightarrow \mathbb{E} g_2[\eta 1_{\eta < 0}] = 0 \Rightarrow \eta 1_{\eta < 0} = 0 \quad \text{P-a.s.}$$

classical comparison theorem Strict $\sim$

$$\Rightarrow \mathbb{E} g_2[\xi | \mathcal{F}_t] \geq \mathbb{E} g_1[\xi | \mathcal{F}_t] \quad \text{P-a.s.}$$

By Thm 4.1 with (A3), we obtain that.

$$\forall s \in [0, T], \forall z \in \mathbb{R}^d, g_1(s, z) \leq g_2(s, z).$$

Lem 4.5

$h: \mathbb{R}^n \rightarrow \mathbb{R}$ convex. bounded from above $\Rightarrow h$ const.

Proof. h convex $\Rightarrow \forall \alpha \in [0,1], h(\alpha y) \leq \alpha h(y) + (1-\alpha)h(0)$

Let $y = x/\alpha \Rightarrow h(x) \leq \alpha h(x/\alpha) + (1-\alpha)h(0)$

h upper bounded $\Rightarrow h(x) \leq \alpha M + (1-\alpha)h(0)$

$\alpha \rightarrow 0^+, h(x) \leq h(0), \forall x \in \mathbb{R}^n$ ①

But $2h(0) \leq h(x) + h(-x) \Rightarrow 2h(0) \leq h(x) + h(0)$
 $\Rightarrow h(x) \geq h(0)$ ②

① + ② $\Rightarrow h(x) = h(0)$ (h is a constant) $\square$

Remark. As a byproduct of this lemma, we deduce that if g satisfies the assumptions (A1) and (A3) and if $(y, z) \mapsto g(s, y, z)$ is convex then in fact g does not depend on y . Indeed, if z is fixed, the function $y \mapsto g(s, y, z)$ is convex. Moreover, since $g(s, y, 0) = 0$, the Lipschitz assumption gives $|g(s, y, z)| \leq K|z|$ and thus by the previous lemma $y \mapsto g(s, y, z)$ is constant. Of course, the same result holds for concave functions. (A1)(A3)凸 $\Rightarrow$ 不依赖于 y .

Corollary 4.6

In Thm 4.4. Assume moreover that P-a.s.

For each $s, z \mapsto g_1(s, z)$ convex (resp. $z \mapsto g_2(s, z)$ concave)

$\forall \xi \in L^2(\mathcal{F}_T), \mathbb{E}_g(\xi) \leq \mathbb{E}_{g_2}(\xi) \Rightarrow \exists (\alpha_t)_{t \in [0,T]}, \text{ p.m. } + b.$

s.t. P-a.s. $\forall s \in [0,T], \forall z \in \mathbb{R}^d, g_1(s, z) = g_2(s, z) = \alpha_s \cdot z$

Proof. By Thm 4.4. We have $g_1(s, z) \leq g_2(s, z)$.

But $z \mapsto (g_1 - g_2)(s, z)$ convex and bounded from above by 0.

By Lem 4.5. $g_1(s, 0) - g_2(s, 0) = 0 \Rightarrow g_1(s, z) = g_2(s, z)$.

$g_1(s, \cdot)$ is convex and concave + $g_1(s, 0) = 0$

$\Rightarrow g_1(s, \cdot)$ is linear and $\exists (\alpha_t)_{t \in [0,T]}, \text{ p.m. } s.t.$

P-a.s. $\forall s \in [0,T], \forall z \in \mathbb{R}^d, g_1(s, z) = g_2(s, z) = \alpha_s \cdot z$.

Moreover $g_1(s, z)$ L in z uniformly w.r.t. s. by (A1)

$\Rightarrow (\alpha_t)_{t \in [0,T]}$ bounded. $\square$

Now assume P-a.s. $z \mapsto g(s, z)$ convex for each s .

Fix $\xi \in L^2(\mathcal{F}_T)$. Consider $F: \mathbb{R} \rightarrow \mathbb{R}$ cvx s.t. $F(\xi)$ square-integrable.

$\partial F(x)$ mean the subgradient of F .

Prop. 5.2

P-a.s. $\partial F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \cap (0,1)^c \neq \emptyset$

Then, P-a.s. $F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \leq \mathbb{E}_g[F(\xi) | \mathcal{F}_t]$.

Proof. $\exists \beta \in \partial F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \cap (0,1)^c, s.t.$

$F(\xi) - F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \geq \beta(\xi - \mathbb{E}_g[\xi | \mathcal{F}_t])$

Set $\Omega_n = \{\beta | \mathbb{E}_g[\xi | \mathcal{F}_t] + |\beta| \leq n\}$. $\beta_n = \beta \mathbb{1}_{\Omega_n}$.

$\mathbb{1}_{\Omega_n} F(\xi) - F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \geq \beta_n(\xi - \mathbb{E}_g[\xi | \mathcal{F}_t])$

$\xi, F(\xi) \in L^2(\mathcal{F}_T)$. β_n bounded by n. F cvx on $C(\mathbb{R})$

so the four terms of above Ineq are square integrable.

$\mathbb{E}_g[\mathbb{1}_{\Omega_n} F(\xi) - F(\mathbb{E}_g[\xi | \mathcal{F}_t]) | \mathcal{F}_t] \geq \mathbb{E}_g[\beta_n(\xi - \mathbb{E}_g[\xi | \mathcal{F}_t]) | \mathcal{F}_t]$

$\beta_n \mathbb{1}_{\Omega_n}$ $\mathcal{F}_t$ -measurable + Lem 4.2 $\Rightarrow$.

$\mathbb{E}_g[\mathbb{1}_{\Omega_n} F(\xi) | \mathcal{F}_t] - \mathbb{1}_{\Omega_n} F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \geq \mathbb{E}_g[\beta_n \xi | \mathcal{F}_t] - \beta_n \mathbb{E}_g[\xi | \mathcal{F}_t]$ (9)

$\beta_n \mathcal{F}_t$ -measurable. $\beta_n \in (0,1)^c$. $\mathbb{E}_g[\beta_n \xi | \mathcal{F}_t] \geq \beta_n \mathbb{E}_g[\xi | \mathcal{F}_t]$

For $t \leq r \leq T$.

$\beta_n Y_r = \beta_n \xi + \int_r^T \beta_n g(s, \beta_n Z_s) ds - \int_r^T \beta_n Z_s \cdot dW_s$

g cvx + Lem 5.1 + $g(s, 0) = 0 \Rightarrow$

$\beta_n Y_r \leq \beta_n \xi + \int_r^T g(s, \beta_n Z_s) ds - \int_r^T \beta_n Z_s \cdot dW_s$

That is, $\beta_n \mathbb{E}_g[\xi | \mathcal{F}_t] \leq \mathbb{E}_g[\beta_n \xi | \mathcal{F}_t]$

Back to (9). $\forall n \in \mathbb{N}^+, \mathbb{1}_{\Omega_n} F(\mathbb{E}_g[\xi | \mathcal{F}_t]) \leq \mathbb{E}_g[\mathbb{1}_{\Omega_n} F(\xi) | \mathcal{F}_t]$.

$n \rightarrow +\infty$ + $\mathbb{E}_g[\cdot | \mathcal{F}_t]$ continuous $\Rightarrow$ Jensen Ineq. $\square$

Remark. It is also possible to obtain the Jensen inequality when the function g is concave instead of being convex. In this context, this requires essentially the fact that the convex function F satisfies $0 \leq F'(x) \leq 1$ (e.g. $F(x) = 1/\pi \{x \arctan x - 1/2 \ln(1+x^2)\} + x/2$).

On Jensen's Inequality.

Lem 5.1

$h: \mathbb{R}^n \rightarrow \mathbb{R}$ convex. $\forall x \in \mathbb{R}$, we have:

$h(\alpha x) \leq \alpha h(x) + (1-\alpha)h(0), \alpha \in [0,1]$

$h(\alpha x) \geq \alpha h(x) + (1-\alpha)h(0), \alpha \in [0,1]^c$.

Proof. Fix $x \in \mathbb{R}^n$. $\forall \alpha \in \mathbb{R}$. Set $f(\alpha) = h(\alpha x) - \alpha h(x) - (1-\alpha)h(0)$
 f convex on $\mathbb{R}$. $f(0) = f(1) = 0 \Rightarrow$ the result $\square$

$$\begin{aligned} f(\beta \alpha_1 + (1-\beta) \alpha_2) &= h(\beta \alpha_1 x + (1-\beta) \alpha_2 x) - \beta \alpha_1 h(x) - (1-\beta) \alpha_2 h(x) \\ &\quad - h(0) + \beta \alpha_1 h(0) + (1-\beta) \alpha_2 h(0) \\ &\leq \beta h(\alpha_1 x) - \beta \alpha_1 h(x) - \beta (1-\alpha_1) h(0) \\ &\quad + (1-\beta) h(\alpha_2 x) - (1-\beta) \alpha_2 h(x) - (1-\beta) (1-\alpha_2) h(0) \\ &= \beta f(\alpha_1) + (1-\beta) f(\alpha_2) \end{aligned}$$

• A Counterexample to Jensen Ineq.



$g: \mathbb{R} \rightarrow \mathbb{R}, z \mapsto z^2$ if $z \in [-1,1]$ and $2|z|-1$ if $|z| > 1$.

Fixed $\sigma \in [0,1], \xi := -\sigma^2 T + \sigma W_T$.

BSDE: $Y_t = \xi + \int_t^T g(r, Z_r) dr - \int_t^T Z_r \cdot dW_r, 0 \leq t \leq T$.

Its solution is $((-\sigma^2 t + \sigma W_t, \sigma))_{t \in [0,T]}$.

$\Rightarrow \frac{1}{2} \mathbb{E}_g[\xi | \mathcal{F}_t] = -\frac{\sigma^2}{2} + \frac{\sigma}{2} W_t$.

But the solution of the following BSDE

$Y_t = \frac{1}{2} \xi + \int_t^T g(r, Z_r) dr - \int_t^T Z_r \cdot dW_r, 0 \leq t \leq T$.

is given by $(\frac{\sigma}{2} W_t - \frac{\sigma^2}{2} T + (\frac{\sigma}{2})^2 (T-t), \frac{\sigma}{2})_{t \in [0,T]}$.

$\Rightarrow \frac{1}{2} \mathbb{E}_g[\xi | \mathcal{F}_t] - \mathbb{E}_g[\frac{1}{2} \xi | \mathcal{F}_t] = (T-t) \frac{\sigma^2}{4} > 0$ ($t < T$).

It contradicts to Jensen Ineq.